

Nama : Syahril Arfian Almazril
Nim : 103032300013
Kelas : IT-47-04

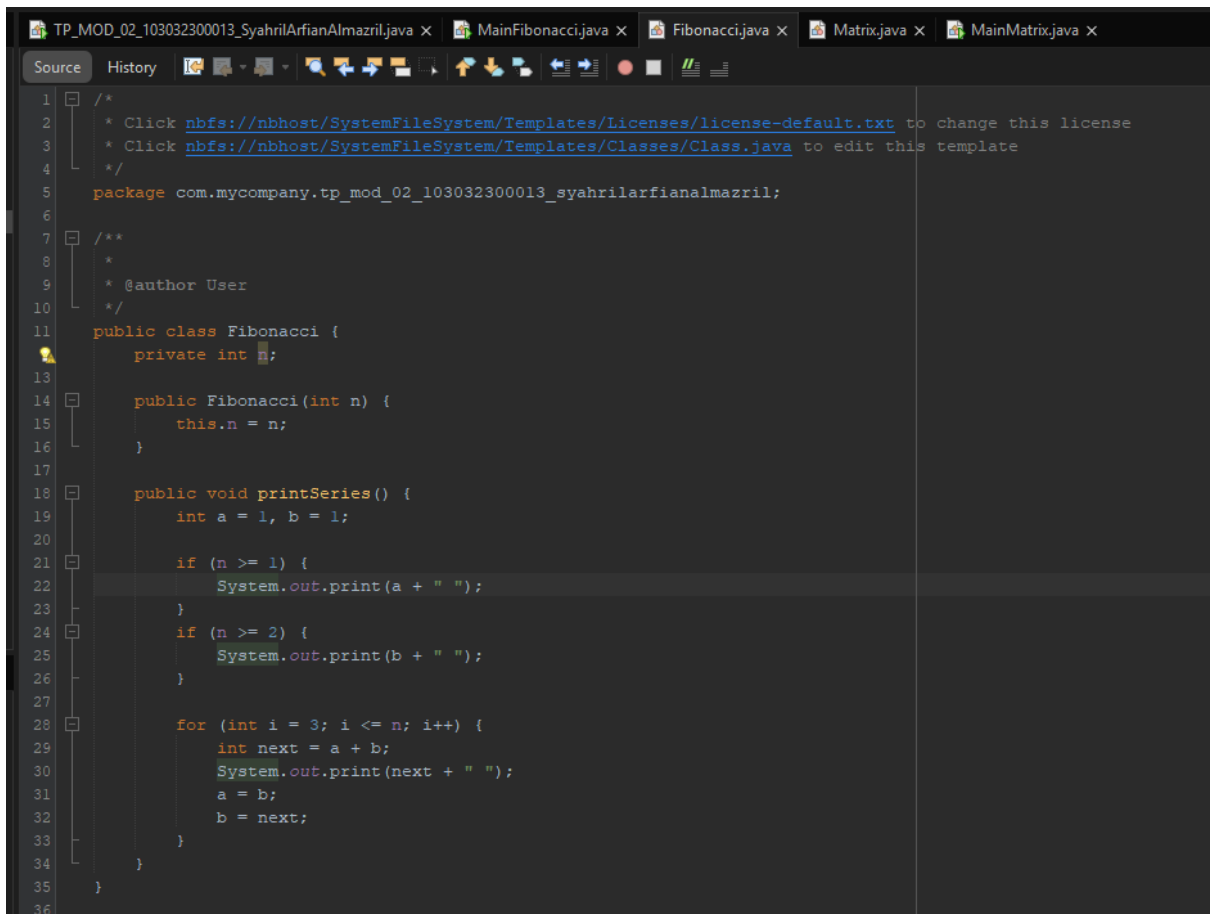
B. Tugas Pendahuluan

1. Buatlah program Java yang menerima sebuah inputan integer $n > 0$, dan meng-outputkan bilangan Fibonacci sampai suku ke- n :

Input	Output
Masukkan n: 6	1 1 2 3 5 8
Masukkan n: 11	1 1 2 3 5 8 13 21 34 55 89

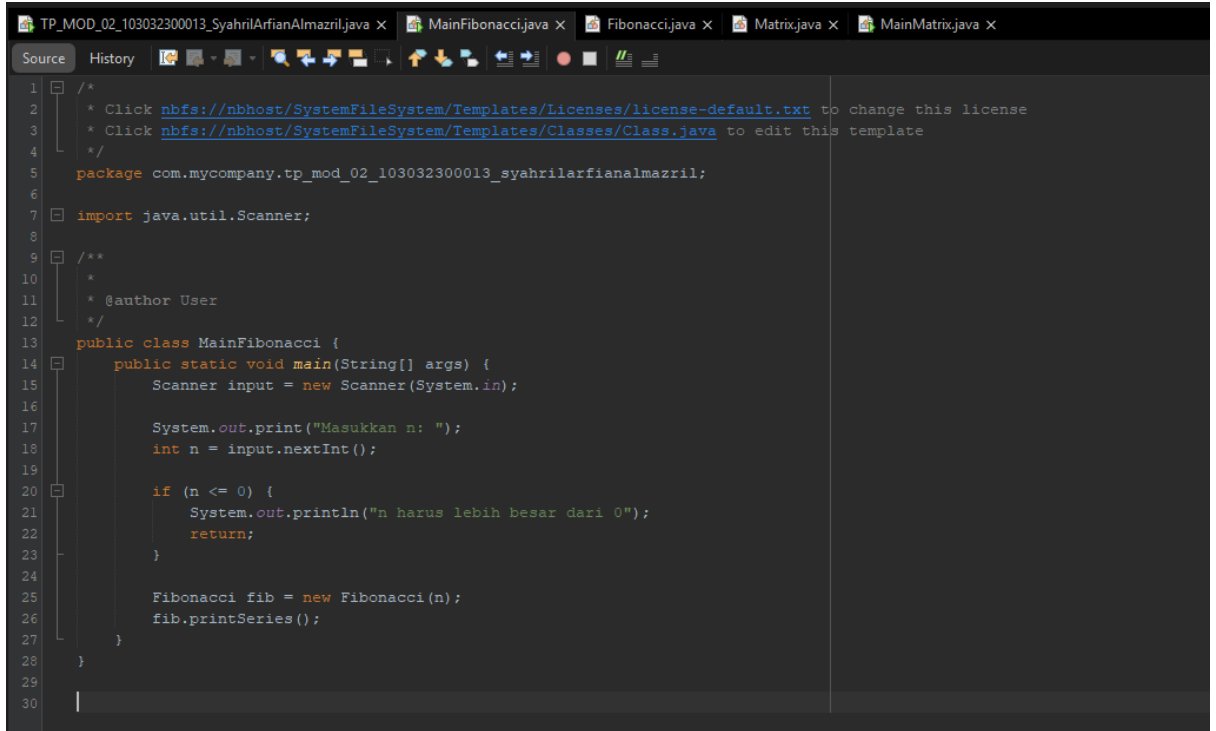
Code :

- Fibonacci.Java



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_02_103032300013_syahrilarfianalmazril;
6
7   /**
8    *
9    * @author User
10   */
11  public class Fibonacci {
12      private int n;
13
14      public Fibonacci(int n) {
15          this.n = n;
16      }
17
18      public void printSeries() {
19          int a = 1, b = 1;
20
21          if (n >= 1) {
22              System.out.print(a + " ");
23          }
24          if (n >= 2) {
25              System.out.print(b + " ");
26          }
27
28          for (int i = 3; i <= n; i++) {
29              int next = a + b;
30              System.out.print(next + " ");
31              a = b;
32              b = next;
33          }
34      }
35  }
```

- MainFibonacci.Java

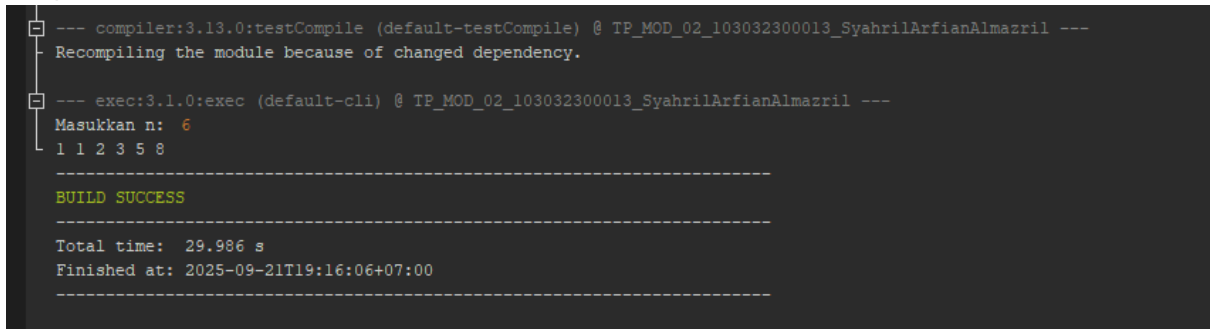


```

1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_02_103032300013_syahrilarfianalmazril;
6
7   import java.util.Scanner;
8
9   /**
10    *
11    * @author User
12    */
13   public class MainFibonacci {
14       public static void main(String[] args) {
15           Scanner input = new Scanner(System.in);
16
17           System.out.print("Masukkan n: ");
18           int n = input.nextInt();
19
20           if (n <= 0) {
21               System.out.println("n harus lebih besar dari 0");
22               return;
23           }
24
25           Fibonacci fib = new Fibonacci(n);
26           fib.printSeries();
27       }
28   }
29
30

```

- Output 1 :



```

--- compiler:3.13.0:testCompile (default-testCompile) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Recompiling the module because of changed dependency.

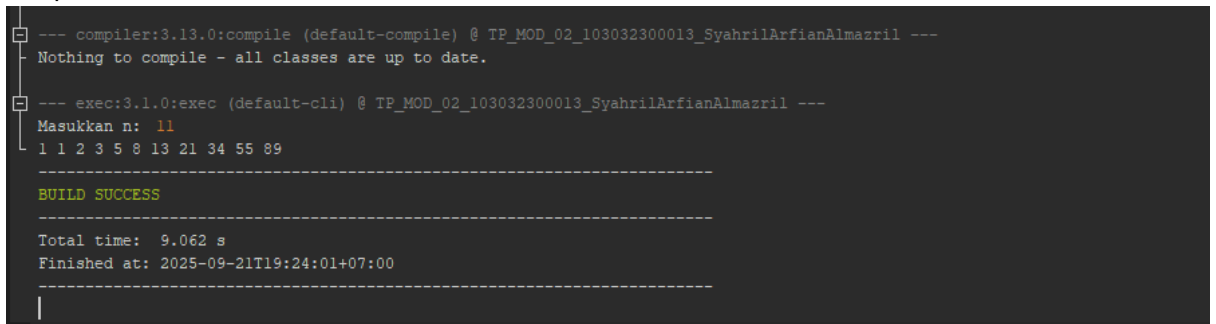
--- exec:3.1.0:exec (default-cli) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Masukkan n: 6
1 1 2 3 5 8

-----
BUILD SUCCESS
-----

Total time: 29.986 s
Finished at: 2025-09-21T19:16:06+07:00
-----

```

- Output 2 :



```

--- compiler:3.13.0:compile (default-compile) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Nothing to compile - all classes are up to date.

--- exec:3.1.0:exec (default-cli) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Masukkan n: 11
1 1 2 3 5 8 13 21 34 55 89

-----
BUILD SUCCESS
-----

Total time: 9.062 s
Finished at: 2025-09-21T19:24:01+07:00
-----

```

- Buatlah program Java yang membuat 2 matrix $n \times n$ dengan n sebagai inputan. Isi tiap nilai matriks (boleh hardcode atau inputan), lalu outputkan 2 matriks tersebut dan matriks hasil perkalian antara 2 matrix tersebut.

Input	Output
Perkalian matriks $n \times n$ n: 2 Isi matriks 1: 3 -2 4 5 Isi matriks 2: 5 1 -1 2	Hasil perkalian: 17 -1 15 14

Code :

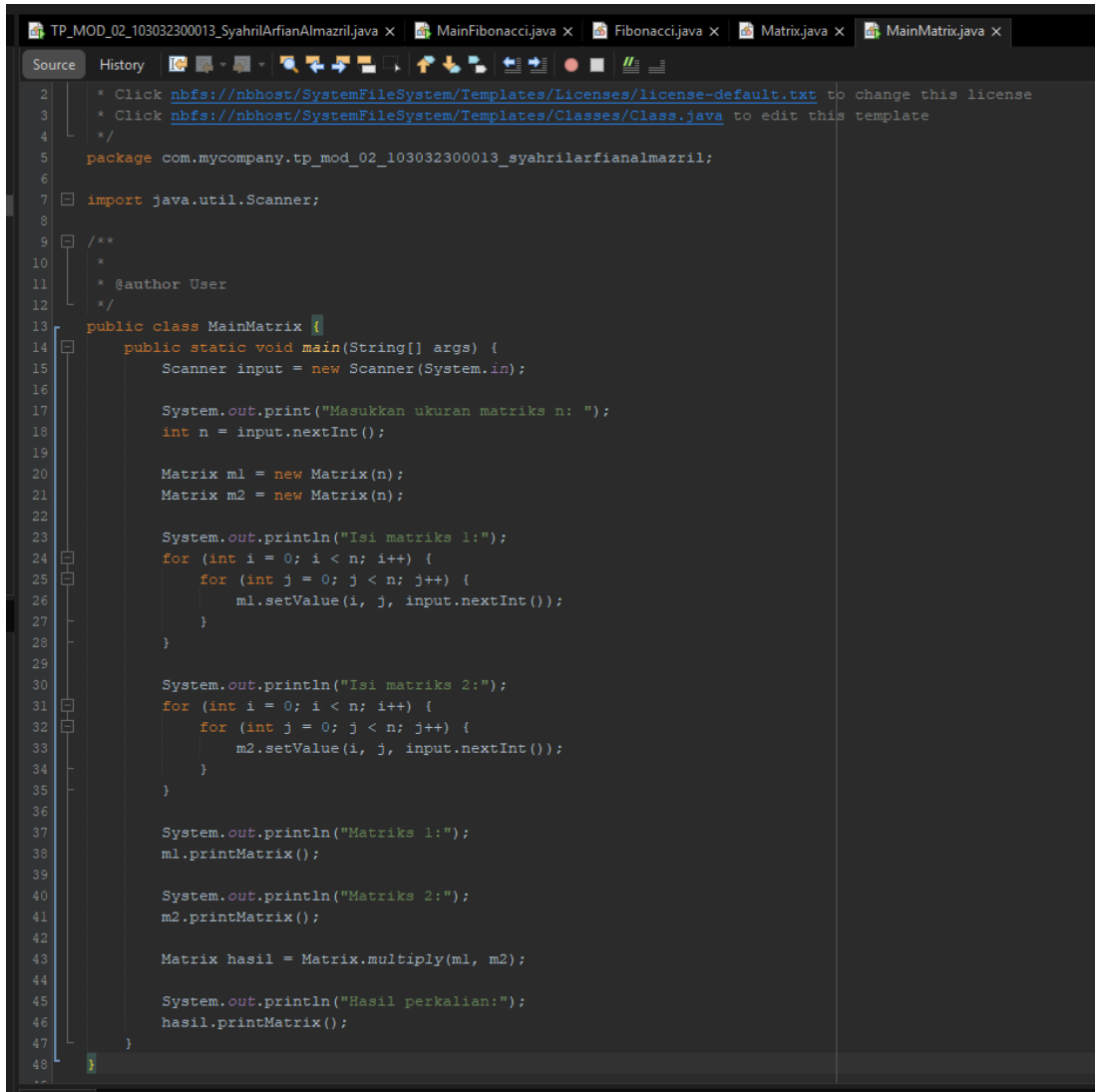
- Matrix.Java

```

5  package com.mycompany.tp_mod_02_103032300013_syahrilarfianalmazril;
6
7  /**
8   *
9   * @author User
10  */
11  public class Matrix {
12      private final int n;
13      private final int[][] data;
14
15      public Matrix(int n) {
16          this.n = n;
17          this.data = new int[n][n];
18      }
19
20      public void setValue(int row, int col, int value) {
21          data[row][col] = value;
22      }
23
24      public int getValue(int row, int col) {
25          return data[row][col];
26      }
27
28      public void printMatrix() {
29          for (int i = 0; i < n; i++) {
30              for (int j = 0; j < n; j++) {
31                  System.out.print(data[i][j] + " ");
32              }
33              System.out.println();
34          }
35      }
36
37      public static Matrix multiply(Matrix a, Matrix b) {
38          int n = a.n;
39          Matrix c = new Matrix(n);
40          for (int i = 0; i < n; i++) {
41              for (int j = 0; j < n; j++) {
42                  int sum = 0;
43                  for (int k = 0; k < n; k++) {
44                      sum += a.getValue(i, k) * b.getValue(k, j);
45                  }
46                  c.setValue(i, j, sum);
47              }
48          }
49          return c;
50      }
51  }

```

- MainMatrix.Java

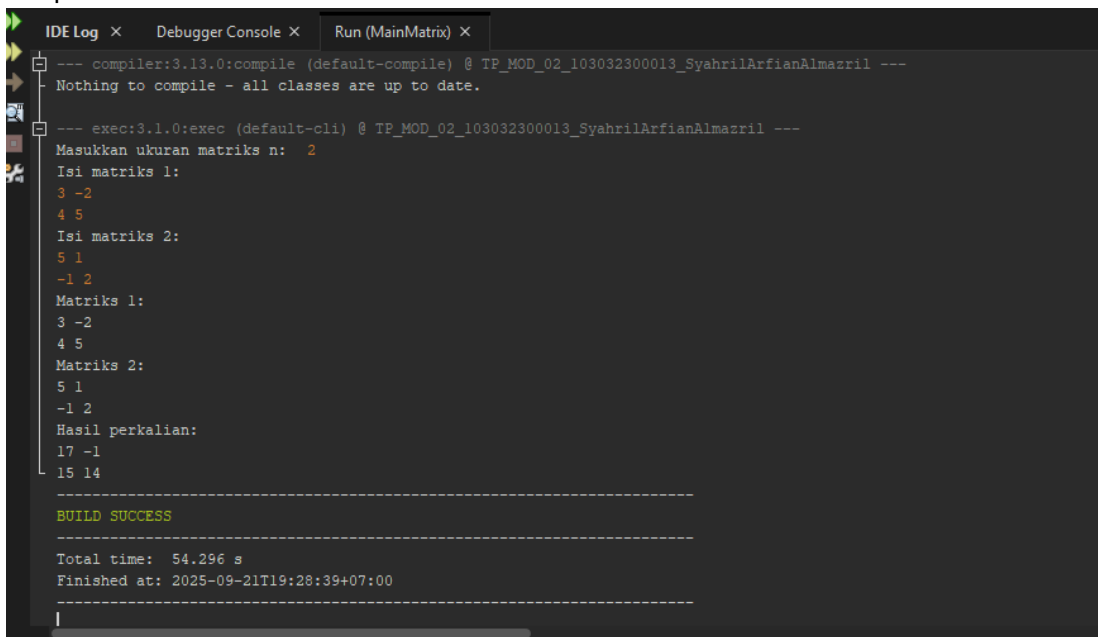


```

2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5  package com.mycompany.tp_mod_02_103032300013_syahrilarfianalmazril;
6
7  import java.util.Scanner;
8
9  /**
10   *
11   * @author User
12   */
13  public class MainMatrix {
14      public static void main(String[] args) {
15          Scanner input = new Scanner(System.in);
16
17          System.out.print("Masukkan ukuran matriks n: ");
18          int n = input.nextInt();
19
20          Matrix m1 = new Matrix(n);
21          Matrix m2 = new Matrix(n);
22
23          System.out.println("Isi matriks 1:");
24          for (int i = 0; i < n; i++) {
25              for (int j = 0; j < n; j++) {
26                  m1.setValue(i, j, input.nextInt());
27              }
28          }
29
30          System.out.println("Isi matriks 2:");
31          for (int i = 0; i < n; i++) {
32              for (int j = 0; j < n; j++) {
33                  m2.setValue(i, j, input.nextInt());
34              }
35          }
36
37          System.out.println("Matriks 1:");
38          m1.printMatrix();
39
40          System.out.println("Matriks 2:");
41          m2.printMatrix();
42
43          Matrix hasil = Matrix.multiply(m1, m2);
44
45          System.out.println("Hasil perkalian:");
46          hasil.printMatrix();
47      }
48  }

```

- Output :



```

--- compiler:3.13.0:compile (default-compile) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Nothing to compile - all classes are up to date.

--- exec:3.1.0:exec (default-cli) @ TP_MOD_02_103032300013_Syahrilarfianalmazril ---
Masukkan ukuran matriks n: 2
Isi matriks 1:
3 -2
4 5
Isi matriks 2:
5 1
-1 2
Matriks 1:
3 -2
4 5
Matriks 2:
5 1
-1 2
Hasil perkalian:
17 -1
15 14

BUILD SUCCESS

Total time: 54.296 s
Finished at: 2025-09-21T19:28:39+07:00

```